MIL-G-13734B(MU)
25 March 1968
SUPERSEDING
MIL-G-13734A(OmIC)
20 June 1958

MILITARY SPECIFICATION

GERMICIDE, QUATERNARY AMMONIUM TYPE

1. SCOPE

1.1 Scope. This specification covers three types of quaternary ammonium germicides for inactivating vegetative types of bacteria.

1.2 Classification. The germicide shall be of the following types as specified (see 6.1):

Type I - Crystalline.

Type II - Paste.

Type III - Liquid.

FSC 6840

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2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-391	-	Plastic, Methacrylate, Sheets, Rods, and Tubes-Cast.
ZZ-G-421	-	Gloves, Rubber, Surgeons'.
PPP-D-723	-	Drums, Fiber.
PPP-D-729	-	Drums: Metal, 55 Gallon (For Shipment of Non-corrosive Materials).
PPP-D-760	-	Drums and Pails, Metal (5 and 16.64 Gallons).
PPP-C-186	-	Container; Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

MILITARY

MIL-P-3803	-	Plastic, Polyethylene, Molded and Extruded Shapes, Sheets, and Tubing.
MIL-T-4239	-	Tape, Pressure-Sensitive Adhesive, Vinyl-Plastic, Opaque, Photographic.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC), Tenth Edition, 1965.

(Application for copies should be addressed to the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, D. C. 20004.)

U. S. PHARMACOPEIA CONVENTION, INC.

The Pharmacopeia of the United States of America (USP) Seventeenth Revision, 1965.

(Copies of the USP may be obtained from the Mack Publishing Company, Easton, Pennsylvania 18042.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Standards

D270 - Sampling Petroleum and Petroleum Products.

D412 - Method of Tension Testing of Vulcanized Rubber.

E70 - Method for Determination of pH of Aqueous Solution.

(Copies of the ASTM Standards may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Chemical composition. The germicide shall be a quaternary ammonium compound and shall contain no mercury or other heavy metal compound, hypochlorite, iodine or other free halogen, formaldehyde, or peroxide. The germicide shall contain not more than three percent, on an active ingredient basis, of impurities which normally occur during the processing of the germicide. Only water may be added.

3.2 Product.

3.2.1 Type I. Type I germicide shall be free-flowing, crystalline powder containing no less than 90 percent of the active ingredient.

3.2.2 Type II. Type II germicide shall be a paste containing no less than 90 percent of the active ingredient.

3.2.3 Type III. Type III germicide shall be an aqueous solution containing no less than 50 percent of the active ingredient.

3.3 Germicide dilution. The contractor shall prepare dilutions of the active ingredient of the germicide and shall certify the highest dilution which will inactivate both Staphylococcus aureus and Salmonella choleraesuis within ten minutes at 20°C when tested as specified in 4.6.2.

3.4 Dilution characteristics. The germicide shall be completely soluble or dispersible in water, and shall form no precipitate or scum when tested as specified in 4.6.3.

3.5 Dermatitis, erythema, edema. The germicide shall cause no greater degree of dermatitis, erythema, or edema than that caused by a saponified cresol solution when tested as specified in 4.6.4.

3.6 pH. The pH of the highest dilution of the germicide prepared as specified in 3.3 shall be no less than 6.0 when tested as specified in 4.6.5.

3.7 Effect on plastics and rubber. The germicide shall show no greater deleterious effect on plastics and rubber than does distilled water when tested as specified in 4.6.6.

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3.8 Compatibility with sodium nitrite.. The germicide shall not form layers, scum, or visible precipitate when mixed with sodium nitrite solution and tested as specified in 4.6.7.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Special provisions.

* 4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification shall govern.

* 4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, and section 5, for which specific inspection has not been provided in the specification have been satisfied.

4.3 Lotting. A lot shall consist of that single, uniform, and homogeneous quantity of germicide of one type produced by one manufacturer, from one formulation, subjected to the same compounding and manufacturing operations, and filled into final containers.

4.4 Sampling.

4.4.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

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4.4.2 For tests.

4.4.2.1 From tank cars. A one-quart all-levels sample shall be taken from each tank car in accordance with ASTM Standard Test Method D270. The sample shall be placed in a clean, dry container and labeled to identify the sample with the tank car represented.

4.4.2.2 From drums, pails, or glass containers. A sample shall be taken from each lot, the size to be calculated on the basis of one-tenth (1/10) of the square root of the number of containers in the lot raised to the next highest whole number. If there are fewer than three containers in a lot, each container shall be sampled. In all other cases, no fewer than three containers shall be selected. From each container in the sample, a representative one-pound specimen shall be taken and placed in separate, clean, dry containers, and labeled to identify the lot and the container represented.

4.5 Inspection provisions.

4.5.1 For examination. Each sample container shall be examined in accordance with the classification of defects and with MIL-STD-105, level S-1.

4.5.2 For tests. Representative samples shall be tested as specified in 4.6.

4.5.3 Classification of defects.

4.5.3.1 Preparation for delivery (level A) (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Packaging materials incorrect or missing where required
102	Container not as specified
103	Container leakage
104	Damaged container
105	Container not lined or coated, when required
106	Marking incorrect, illegible, or missing

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4.6 Tests. Tests shall be conducted as follows.

4.6.1 General. Use analytical reagent and biological grade chemicals and distilled water throughout the tests. Prepare an aqueous solution of each composite germicide sample in the dilution certified by the contractor in a volume sufficient to perform all of the tests specified herein (approximately 1 liter), using recently boiled distilled water at room temperature, and protecting the diluted germicide from atmospheric carbon dioxide. Mix the dilution of the germicide for 30 minutes by means of mechanical shaking or stirring. In addition, prepare a 25 percent aqueous solution, if not one of the above dilutions certified by the contractor, in the same manner specified above. Test this 25 percent solution in addition to the above certified dilutions for dilution characteristics.

4.6.2 Germicidal properties.

4.6.2.1 Resistance of test organism.

4.6.2.1.1 Staphylococcus aureus. Use Staphylococcus aureus, which is inactivated at 20°C by a 1 to 60 dilution of USP phenol between 5 and 10 minutes, and which is not inactivated by a 1 to 70 dilution of USP phenol in 10 minutes using procedures specified in chapter 5 of the Official Methods of Analysis of the Association of Official Analytical Chemists.

4.6.2.1.2 Salmonella choleraesuis. Use Salmonella choleraesuis in a 48 to 54 hour culture, which is inactivated at 20°C by a 1 to 90 dilution of USP phenol in 10 minutes, and which is not inactivated by a 1 to 100 dilution of USP phenol in 10 minutes using procedures specified in chapter 5 of the Official Methods of Analysis of the Association of Official Analytical Chemists.

4.6.2.2 Procedure. Determine the germicidal properties of each lot, using both Staphylococcus aureus and Salmonella choleraesuis as test organisms, employing the Use-Dilution Method-Final Action specified in Official Methods of Analysis of the Association of Official Analytical Chemists, testing the certified dilutions of the germicide (see 4.6.1) as the use-dilution specified in the procedure.

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4.6.3 Dilution characteristics. Examine the dilutions of the germicide (see 4.6.1) visually for precipitated matter and scum. Set aside two 100-milliliter portions of each dilution, maintaining one set at a temperature of 70° ± 2°C and the other set at a temperature of 10° ± 1°C for a period of two weeks. At the end of the exposure period, visually examine the contents of the cylinders for evidence of a precipitate or scum.

4.6.4 Dermatitis, erythema, edema. Closely clip the hair on both sides of five healthy albino rabbits which have had no previous toxicological exposure and which weigh between 2 and 3.5 kilograms each. (Do not remove the hair by shaving or depilatories.) Prepare 20 pads of cotton gauze two layers thick and approximately one-inch square by saturating 10 pads with the dilution of the germicide (see 4.6.1) and 10 with a 2.0 percent saponified cresol solution in distilled water. Immediately tape the pads to the clipped sides of the rabbits, placing one dilution of the germicide and one cresol solution saturated pad on each side of each rabbit, using waterproof tape to minimize evaporation. Remove the pads after eight hours exposure. Examine the test areas immediately and at 16 and 24-hour intervals after removal. Repeat this test twice using the same rabbits (clipping the hair if necessary), with each test beginning 48 hours after the beginning of the previous test.

4.6.5 pH. Place 100 ml of the dilution of the germicide (see 4.6.1) in a clean glass beaker. Determine the pH of the * dilution, using the Standard Method of Test for pH of Aqueous Solutions with the Glass Electrode, ASTM Designation E70.

4.6.6 Effect on plastics and rubber. Prepare three, two-inch squares each of methacrylate 0.060 inch nominal thickness, * conforming to type I of L-P-391, and of rubber from a glove conforming to grade B of ZZ-G-421. Immerse one square of each material in water and one square of each material in an aqueous solution of the germicide containing one percent of the active ingredient. Allow the squares to remain immersed in the solution for a period of 48 hours at a temperature of 70° ± 20°C. At the end of the immersion period, remove the squares, wash them with * water, and allow them to dry at room temperature. At the end of the drying period test the sample immersed in water for elongation and set in accordance with ASTM Method D412. Test results obtained shall be the standard for measurement of the test samples immersed in the germicidal solution. Perform a similar test on the samples immersed in the germicide solution.

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* 4.6.7 Compatibility with sodium nitrite. Prepare a 50 percent aqueous solution of sodium nitrite, applying heat if necessary, to facilitate solution. Cool the solution to room ambient and add an equal volume of 50 percent germicide. Examine the resultant mixture for evidence of separation, scum, or precipitate.

4.7 Rejection criteria. If any sample fails one or more of the above tests, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A.

5.1.1.1 Unit packaging. The germicide shall be packaged in unit quantities up to but not exceeding one-gallon capacity glass containers in accordance with the applicable requirements of PPP-C-186. All closures shall be furnished with inner seals. The closures shall be further sealed with pressure-sensitive tape, conforming to MIL-T-4239, and of sufficient width to extend well down on the neck or body of the container and slightly over the top edge of the cap. The tape shall adhere closely to the cap and to the neck or body of the container.

5.1.1.2 Intermediate packaging. The germicide, packaged as specified in 5.1.1.1, shall be intermediately packaged in accordance with the applicable requirements of PPP-C-186.

5.1.2 Level B. Not applicable.

5.1.3 Level C. The germicide shall be packaged in unit quantities up to but not exceeding one-gallon capacity containers and in accordance with the manufacturer's commercial practice, when adequate to provide protection for the items packed as specified in 5.2.3.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Unit quantities not exceeding one gallon. Unit quantities packaged as specified in 5.1.1 shall be packed in conformance with the applicable requirements of PPP-C-186.

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5.2.1.2 Unit quantities exceeding one gallon. The germicide shall be packed in a five-gallon container conforming to type I or II, class 2, of PPP-D-760, or in a 55-gallon drum conforming to type III, grade A, of PPP-D-729 or PPP-D-723, as applicable. Metal or fiber containers shall be lined with a minimum thickness of 0.0030 polyethylene conforming to grade I of MIL-P-3803, or coated with a polyethylene material that will not be affected by the germicide nor have a deleterious effect on the germicide.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The germicide, packaged as specified in 5.1.3, and unit quantities in excess of one gallon capacity, including bulk quantities, shall be packed to afford protection against damage during direct shipment from the supply source to the first destination for immediate use. Containers shall comply with I.C.C. Regulations, Uniform Freight Classification Regulations, or other common carrier regulations applicable to the mode of transportation.

* 5.3 Marking. In addition to any special marking required by the contract or order; unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129. The item identification shall include the "certified" highest dilution, specified by the supplier, which will inactivate both Staphylococcus aureus and Salmonella choleraesuis. Containers shall be marked with any special marking required by the Interstate Commerce Regulations for shipment of germicides.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of germicide required (see 1.2 and 3.2).

(c) Unit quantity and type of unit package.

(d) Level of packaging and packing.

(e) That the bidder forward the certification specified in 3.3 for evaluation with his bid.

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6.2 Evaluation of the germicide. The certified highest dilution of the germicide which will inactivate micro-organisms should be used, for procurement purposes, as an aid in the evaluation of the germicide.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 6840-A196

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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